



M002 Week 5 — Mini Cube Puzzle

Topic: Mini Cube Puzzle · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

Finish the mini cube. The starter builds part of it. Keep it going: add **2 more conditionals** with **one OR**, fill the cube, then break a hole in the **top** and climb out.

1 · Predict 🧠

Read each block. Write what you think happens.

```
for i in range(5):
    agent.place(BLOCK)
    agent.move(FORWARD, 1)
```

What shape is left behind? How long is it?


```
if on_edge or on_top_layer:
    agent.place(BLOCK)
else:
    pass # leave empty
```

Only one condition must be true. Which spots get a block? Which stay empty?

2 · Trace 🦶

```
for layer in range(3):
    fill_layer()
    agent.move(UP, 1)
if agent.detect(BLOCK, UP):
    # break above, then climb
    agent.move(UP, 1)
```

Walk through it. How many layers fill? Where does the agent end up? Is there still a block above it?

3 · Spot + Fix the Bugs 🐛

Bug A — The agent should place blocks across the layer, **then** move up. It moves up too early.

```
agent.place(BLOCK)
agent.move(UP, 1)
agent.place(BLOCK)
agent.place(BLOCK)
```

Fixed code:

Why was it wrong? Why does your fix work?

Bug B — Place a block when on the **bottom layer** **OR** on an **outer wall**. It only checks the bottom.

```
if on_bottom_layer:
    agent.place(BLOCK)
```

Hint: add the wall check with **or**.

Fixed code:

Why was it wrong? Why does your fix work?

Bug C — To climb out, break the block above **first**, then move up. Right now it moves up into solid block and gets stuck.

```
agent.move(UP, 1)
agent.destroy(UP)
```

Hint: make the hole, then move into it.

Fixed code:

Why was it wrong? Why does your fix work?

4 · Write Code 📌

Write a loop that fills **one layer**: place a block, move forward, repeat 4 times.

Write a conditional that uses **or** : if the agent is on a **corner** **OR** on the **top**, place a block.

mini cube starter

- or

4 to 6 sentences:

First, I looked at the starter and saw ...

To keep building, I ...

I used a conditional when ...

I made the hole in the top by ...

One tricky moment was ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Film the agent building on a phone. Teach it like the viewer has never seen it. Use these words: **cube**, **layer**, **conditional**, **OR**, **hole**.

1. Show the starter cube before you run.
2. Run your code and watch the cube fill.
3. Read your conditional out loud and say what makes it true.
4. Show the agent breaking the hole in the top and climbing out.

Write what you will say. Plan it before you film — you can read from it.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.